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Emergency communications testing using short codes

Abstract

Systems and methods for performing emergency communications testing are disclosed. Network components, such as a text control center, may be configured to recognize a short code in a text message and enter an emergency communications testing mode for the user device that transmitted the short code. In testing mode, the network components may process emergency communications by determining the emergency processing data (e.g., routing instructions, appropriate PSAP, etc.) and providing that information to the user device rather than forwarding such communications to an emergency services provider. The network component may revert to normal operation after a period of time or after processing a predetermining number of emergency communications.

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Background/Summary

BACKGROUND

(1) The number of wireless communications devices in use has grown exponentially, along with the number of communications networks used to support such devices. Typical (e.g., non-emergency) communications facilitated by such networks do not include an indication of a current wireless communications device location. For example, while wireless communications devices typically include location determination components and/or services, such as global positioning system (GPS) components, the current geographical location of a wireless communications device is not indicated in its communications with a network and/or other devices to protect user privacy. A location may be associated with a device identifier, such as a telephone number (e.g., based on area code) that may be included or determined for the device's communications, but this location remains the same for the identifier regardless of the actual current geographical location of the wireless communications device and therefore is not an accurate indicator of the device's current location.

(2) Emergency communications are an exception to this general practice of maintaining current user device location privacy. In communications to an emergency services agency or emergency services personnel (e.g., police ambulance, 911, etc.), the device's location may be key to quickly locating the user of the device and providing assistance. Therefore, in emergency communications (e.g., calls or texts to 911), a wireless communications device may be configured to include a current device location as determined by its location determinations components and/or services. While this configuration protects user privacy while facilitating a proper response in emergency situations, it makes testing new or updated devices and network configurations for proper emergency messaging and behavior challenging because emergency communications are

necessarily transmitted to emergency services systems. It is currently difficult to test the processing of emergency communications without involving emergency services in a non-emergency or without using test messages that are not truly reflective of emergency messages and therefore may not adequately test such processing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is described with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The same reference numbers in different figures indicate similar or identical items.

(2) FIG. **1** is a schematic diagram of an illustrative wireless communication network environment in which systems and techniques for testing emergency communications using short codes may be implemented, in accordance with examples of the disclosure.

(3) FIG. **2** is a diagram of an illustrative signal flow associated with systems and techniques for testing emergency text communications using short codes, in accordance with examples of the disclosure.

(4) FIG. **3** is a diagram of an illustrative signal flow associated with systems and techniques for testing emergency voice communications using short codes, in accordance with examples of the disclosure.

(5) FIG. **4** is a flow diagram of an illustrative process for performing testing of emergency text communications using short codes, in accordance with examples of the disclosure.

(6) FIG. **5** is a flow diagram of an illustrative process for performing testing of emergency voice communications using short codes, in accordance with examples of the disclosure.

(7) FIG. **6** is a flow diagram of an illustrative process for performing testing of emergency voice communications using short codes, in accordance with examples of the disclosure.

(8) FIG. **7** is a schematic diagram of illustrative components in an example user device that is configured for testing emergency communications using short codes, in accordance with examples of the disclosure.

(9) FIG. **8** is a schematic diagram of illustrative components in an example computing device that is configured for testing emergency communications using short codes, in accordance with examples of the disclosure.

DETAILED DESCRIPTION

Overview

(10) This disclosure is directed in part to systems and techniques for more efficiently and accurately testing the processing of emergency communications of various types that may be initiated by a user equipment (UE) (e.g., smartphone, cell phone, mobile device, wireless communication device, mobile station, etc.) in advanced wireless communications networks. Such advanced networks include networks that support one or more 3GPP standards, including, but not limited to, Long Term Evolution (LTE) networks (e.g., 4G LTE networks) and New Radio (NR) networks (e.g., 5G NR networks). However, the disclosed systems and techniques may be applicable in any network or system in which a user device may request and receive access to communicate with emergency services and/or systems using any protocol.

(11) Based on current regulations and requirements, emergency communications are routed based on the current location of the originating device. For example, a 911 call or a text to 911 initiated by a UE may be routed to a public safety access point (PSAP) that is the closest (e.g., geographically most proximate) PSAP to the current location of the UE from among available PSAPs. A PSAP may be configured to receive such emergency communications and provide them to one or more human operators and/or emergency services systems associated with a particular

geographical area. The emergency communications received by the PSAP may include an indication of a current geographical location of the originating device (e.g., UE). This location may be determined by the UE and/or one or more systems within the network supporting the UE for inclusion in the emergency communications. For example, a UE may include location determination components and/or services, such as GPS components, that may determine a current geographical location of a wireless communications device that may be included in an emergency communication initiated by the UE.

(12) Various operations may be performed within a wireless communications network in order to determine a correct PSAP for an emergency communication and to provide the emergency communication and relevant data to the PSAP. These operations may involve various components within the wireless communications network that may be added, removed, and/or modified similarly to any other components of such networks. The operations performed in processing emergency communications may also be changed over time for various reasons, regardless of whether the components performing such operations change. Therefore, it is important to test such operations and components when they are adjusted and/or from time to time to ensure the emergency communications are processed properly. This is especially important for emergency communications processing operations and components because such communications are used in situations where the health, safety, and even life of a user may be at stake.

(13) Properly processed emergency communications are transmitted to a PSAP that is most geographically proximate to the originating device to ensure the most rapid response from first responders and to assist in emergency response activities. This makes testing emergency communications processing without actually involving emergency services personnel and systems difficult. In current systems and techniques, a designated alternative number may be used for emergency communications testing (e.g., other than 911). For example, a UE may be operated to transmit texts or originate voice calls to “912,” “919,” or some other alternative testing destination number. However, many UEs are configured to include current UE location information only with legitimate emergency communications (e.g., calls and texts to 911), excluding the current user location from any other type of communication to protect user privacy. Due to the wide variety of UEs and other user devices that may be supported by a wireless communications network and the proprietary nature of the components and software configured on such devices, it is not practical, and may not even be possible, for a network operator to modify this behavior in UEs for testing purposes. Moreover, there is no standardized or commonly agreed upon testing number(s), which prevents manufacturers of UEs and other wireless devices from configuring such devices in advance to include a current device location in testing communications.

(14) The disclosed systems and techniques may be used to test emergency communications more accurately and efficiently without requiring the alteration of UEs or other user devices and without involving actual emergency services systems and personnel in the testing process. In various examples, a network component, such as a text control center (TCC) or a gateway mobile location center (GMLC), may be configured to initiate an emergency communications testing mode for a particular user device (e.g., UE). While in an emergency communications testing mode, this component may alter its normal behavior for testing emergency communications and/or to instruct one or more other network components to alter their respective behaviors for testing purposes. These behavior operations may be implemented in response to receiving one or more particular text messages from a UE or other user device operated by a testing user. The TCC and/or the one or more altered network components may revert to normal behavior after meeting one or more criteria, receiving one or more notifications, and/or completing one or more testing operations.

(15) For example, as described in more detail herein, a TCC may be configured to enter an emergency text communications testing mode (“text testing mode”) in response to receiving a particular text short code from a UE. Such short codes may take any form and may include any number and combination of alphanumeric characters. For instance, a text short code of “922” may

be used herein as an exemplary short code initiating a text testing mode without limitation. In response to receipt of a text texting mode short code, a TCC may enter a text testing mode for communications originating with the UE that originated the text testing mode short code. Further in response to such a text texting mode short code, the TCC may reply to the originating UE with a testing mode confirmation communication that may include one or more parameters, details, and/or other information associated with the text testing mode.

(16) While in text testing mode for a particular UE, the TCC may perform testing operations in response to receiving an emergency text communication from that UE, for example, instead of performing normal emergency communications processing operations. For example, a testing operator that operated the UE to transmit the text texting mode short code (and resultingly place the TCC in text testing mode) may then operate the UE to transmit an emergency text (e.g., to 911). In response to receiving this emergency text, the TCC, because it is in text testing mode, may then determine the UE's current location (e.g., based on the emergency text). The TCC may next determine the appropriate PSAP for the UE's current location. Rather than transmitting the emergency text to that PSAP, because the TCC is in text testing mode for this UE, the TCC may reply to the UE with a testing information communication that may include the determined PSAP, the determined current UE location, a timestamp, additional PSAP data, and/or other testing information.

(17) In various examples, the TCC may exit the text testing mode or otherwise return to normal text operating mode after responding to a threshold number of emergency texts. For example, the TCC may be configured to remain in text testing mode until one emergency text has been processed for the associated UE. Alternatively or additionally, the TCC may exit the text testing mode or otherwise return to normal text operating mode after a threshold period of time has passed, in some examples, regardless of the number of emergency texts processed in text testing mode. For example, the TCC may be configured to remain in text testing mode for 5 minutes, 10 minutes, 1 hour, etc. Alternatively or additionally, the TCC may exit the text testing mode or otherwise return to normal text operating mode after receiving another short code from the originating UE. For example, the TCC may be configured to exit text testing mode in response to receiving a “922x” or “822” short code from the originating UE. Other criteria may be used in addition, or instead, by a TCC to determine when to revert to normal operation from text testing mode for a particular UE. Note that any such criteria may be indicated in a testing mode confirmation communication and/or in a testing information communication.

(18) Once back in normal operating mode, a TCC may handle subsequent emergency texts from the UE formally in text testing mode normally. That is, the TCC will, in response to receiving an emergency text from the UE, determine the current UE location and the appropriate PSAP for that location and forward the emergency text to the PSAP. The TCC may also, or instead, further facilitate emergency communications between the UE and the PSAP.

(19) The disclosed systems and techniques may also be used to test emergency voice communications. For example, as described in more detail herein, a gateway mobile location center (GMLC) may be configured to enter an emergency voice communications testing mode (“voice testing mode”) in response to receiving a particular text short code from a UE (e.g., via short message service center (SMSC)). Such short codes may take any form and may include any number and combination of alphanumeric characters. For instance, a text short code of “933” may be used herein as an exemplary short code initiating a voice testing mode without limitation. In response to receipt of a voice texting mode short code, a GMLC may enter a voice testing mode for communications originating with the UE that originated the voice testing mode short code. In response to such a voice texting mode short code in a text, the SMSC may forward the short code to a network location-based services component, such as a GMLC, to enter a voice testing mode. The location-based services component may reply with a confirmation message that may include voice testing mode data. The SMSC may forward this confirmation message to the originating UE

with a testing mode confirmation communication that may include one or more parameters, details, and/or other information associated with the voice testing mode.
(20) While in voice testing mode for a particular UE, the location-based services component may perform testing operations in response to receiving an emergency voice communication from that UE, for example, instead of performing normal emergency communications processing operations. For example, a testing operator that operated the UE to transmit the voice testing mode short code (and resultingly place the GMLC and/or other location-based services component in voice testing mode) may then operate the UE to initiate an emergency voice call (e.g., a call to 911). In response to receiving a request to initiate this call, the GMLC and/or other location-based services component, because it is in voice testing mode, may then determine the UE's current location (e.g., based on the emergency text). The GMLC and/or other location-based services component may next determine the appropriate PSAP for the UE's current location. Rather than setting up the emergency call between the UE and the PSAP, because the GMLC and/or other location-based services component is in voice testing mode for this UE, the GMLC and/or other location-based services component may set up a testing information communication (e.g., a voice communication and/or a text communication) between the UE and a PSAP simulation function that may indicate the determined PSAP, the determined current UE location, a timestamp, additional PSAP data, and/or other testing information.
(21) In various examples, the GMLC and/or the location-based services component may exit the voice testing mode or otherwise return to normal voice operating mode after responding to a threshold number of emergency call requests. For example, the location-based services component may be configured to remain in voice testing mode until one emergency call has been processed for the associated UE. Alternatively or additionally, the GMLC and/or other location-based services component may exit the voice testing mode or otherwise return to normal voice operating mode after a threshold period of time has passed, in some examples, regardless of the number of emergency calls processed in voice testing mode. For example, the GMLC and/or other location-based services component may be configured to remain in voice testing mode for 5 minutes, 10 minutes, 1 hour, etc. Alternatively or additionally, the GMLC and/or other location-based services component may exit the voice testing mode or otherwise return to normal voice operating mode after receiving another instruction via the SMSC that may include a short code from the originating UE. For example, the SMSC may be configured to transmit to the GMLC and/or location-based services component an exit voice testing mode message that may include a “933x” or “833” short code from the originating UE. Other criteria may be used in addition, or instead, by a GMLC and/or a location-based services component to determine when to revert to normal operation from voice testing mode for a particular UE. Note that any such criteria may be indicated in a testing mode confirmation communication and/or in a testing information communication.
(22) Once back in normal operating mode, a GMLC and/or location-based services component may handle subsequent emergency calls from the UE formally in voice testing mode normally. That is, the location-based services component will, in response to receiving an emergency call request from the UE, determine the current UE location and the appropriate PSAP for that location and provide routing instructions for the emergency call to one or more nodes in the network to facilitate the emergency call between the UE and the PSAP. An SMSC may not be involved in such normal emergency call operations as such operations may not involve text communications.
(23) By facilitating the more efficient and accurate testing of emergency communications operations and components, systems and methods described herein can improve the performance and increase the efficiency of network resources (and therefore UE resources), while improving the user experience by ensuring that emergency communications are properly processed. For example, the methods and systems described herein may be more efficient and/or more robust than conventional techniques, as they may increase the efficiency of UE and network resource utilization by facilitating the testing of communications truly reflecting actual emergency

communications without interfering with operational emergency services or requiring modifications to user equipment. Moreover, the methods and systems described herein provide a technological improvement over existing emergency communications testing systems and processes by facilitating an improved testing user experience and increasing network efficiency by reducing the utilization of operational emergency communications resources communications when testing emergency communication operations and components. In addition to improving the efficiency of network and device resource utilization, the systems and methods described herein can provide more robust systems by, for example, making more efficient use of network devices and user devices by reducing unnecessary and/or unproductive device and network modifications and testing signaling, thereby freeing network and user device resources for more productive operations.

(24) Illustrative environments, signal flows, and techniques for implementing systems and techniques for testing emergency communications using short codes are described below. However, the described systems and techniques may be implemented in other environments.

(25) Illustrative System Architecture

(26) FIG. **1** is a schematic diagram of an illustrative wireless network environment **100** in which the disclosed systems and techniques may be implemented. The environment **100** may include a UE **110** that may wirelessly communicate with a base station **120**. While referred to as a “base station” for explanatory purposes herein, the base station **120** may be any type of base station, including, but not limited to, any type of base transceiver station (BTS), NodeB, eNodeB, gNodeB, etc. The base station **120** may communicate with other devices and elements in the core of a wireless communications network **101**. The environment **100** may also include a UE **112** that may wirelessly communicate with a base station **122** that may also be any type of base station, including, but not limited to, any type of BTS, NodeB, eNodeB, gNodeB, etc. The base station **122** may also communicate with other devices and elements in the core of a wireless communications network **101**. The wireless communications network **101** may be any one or more networks that facilitate communications between devices of various types, such as computing devices and mobile devices (e.g., UEs). Various connections between devices in the network **101** may be wired, wireless, or a combination thereof. In various embodiments, the wireless communications network **101** may facilitate communications with one or more wireless devices, such as UEs. The wireless communications network **101** may facilitate packet-based communications between such wireless devices and devices on the Internet and/or one or more systems and devices external to the wireless communications network **101**, such as PSAP **170** and PSAP **172**.

(27) The UE **110** and the UE **112** may exchange text communications with a short message service center (SMSC) **130** via the base station **120** and the base station **122**, respectively. The SMSC **130** may facilitate communications between these UEs and a TCC **140**, that may represent one or more TCCs. For example, the SMSC **130** may direct text communications from the UEs to the TCC **140** for further processing, and relay communications received from the TCC **140** to the UEs, as described in more detail herein. The SMSC **130** may also, or instead, direct text communications from the UEs to other entities for further processing, such as one or more call routing nodes **150** and/or a GMLC **160**, and relay communications received from such entities to the UEs, as described in more detail herein.

(28) The wireless communications network **101** may further include one or more call routing nodes **150** that may be any one or more routing nodes of any type configured to route or otherwise facilitate voice calls. The UE **110** and the UE **112** may exchange call establishment data and voice communications with the call routing nodes **150** via the base station **120** and the base station **122**, respectively. The call routing nodes **150** may interact with the GMLC **160** to determine call routing instructions based on, for example, UE locations. For example, the GMLC **160** may determine an appropriate PSAP for a UE based on the UE's current location. In some examples, the GMLC **160** may be dedicated to providing location-based services for emergency communications, while in

other examples, the GMLC **160** may also provide other location-based services.
(29) The wireless communications network **101** may additionally facilitate communications between the UEs **110** and **112** and external systems, such as the PSAP **170** and/or the PSAP **172**. For example, the UEs may exchange text communications with the PSAPs via the gNodeBs, SMSC **130**, and/or TCC **140**. The UEs may establish voice communications with the PSAPs via the gNodeBs and call routing nodes **150**, using location determination services of the GMLC **160**.
(30) Note that the components, systems, services, and functions represented in the environment **100** are an exemplary subset of components, systems, services, and functions that may be configured in a wireless communications environment. One skilled in the art will recognize that many other components, systems, services, and functions may be configured in such an environment and interact with the components, systems, services, and functions represented in FIG. **1**.
(31) Various emergency communications operations in the wireless communications network **101** may be tested using one or both of the UE **110** and UE **112** according to the disclosed examples. The various signal paths associated with setting up and performing such testing as well as performing actual emergency communications are illustrated in FIG. **1**.
(32) For example, the UE **110** may initiate a text testing operation by transmitting a text testing request to the TCC **140** via the base station **120** and the SMSC **130**. The signal path **181** may represent the path that such a request and related signals may take. The text testing request from the UE **110** may include a short code (e.g., "922") that may cause the TCC **140** to enter an emergency text testing mode for the UE **110**. In response to receiving such a text texting request and entering a text testing mode for the UE **110**, the TCC **140** may reply to the UE **110** with a testing mode confirmation communication. This confirmation may indicate a confirmation of text testing mode at the TCC **140** for the UE **110** and/or one or more parameters, details, and/or other information associated with the text testing mode. For example, the confirmation may explicitly indicate that the testing mode will be active for a particular number of test emergency texts, a particular amount of time, until another (same or different) short code is received, etc. The TCC **140** may store an indicator of text testing mode and an indicator of the UE **110** to maintain a record of the testing mode for the UE **110**. The indicator of the UE **110** may take any form, including a telephone number, a mobile station international subscriber directory number (MSISDN), an international mobile subscriber identity (IMSI), a mobile device number (MDN), etc. For example, the TCC **140** may store an array of identifiers associated with UEs for which the TCC **140** may be in text testing mode.
(33) While in text testing mode for the UE **110**, the TCC **140** may perform testing operations in response to receiving an emergency text communication from the UE **110**. For example, an operator of the UE **110** may transmit a test emergency text (e.g., to 911) that may be received by the TCC **140** via the base station **120** and the SMSC **130**. The signal path **183** may represent the path that such a test emergency text and related signals may take. In response to receiving this emergency text, the TCC **140** may then determine that the text originated with the UE **110**, determine that the TCC **140** is in text testing mode for the UE **110** (e.g., by comparing an identifier in the test emergency text to stored UE identifiers associated with UEs for which the TCC is in text testing mode), and responsively determine a current location for the UE **110** (e.g., based on location data include by the UE **110** in the test emergency text). The TCC **140** may next determine the appropriate PSAP for the UE **110**'s current location (e.g., using the GMLC **160** (not included in the signal path **183**) and/or other means). The TCC **140** may then transmit a testing response to the UE **110** with testing information, such as one or more of the determined PSAP, the determined current UE location, a timestamp, additional PSAP data, and any other testing information.
(34) The TCC **140** may exit the text testing mode or otherwise return to normal text operating mode based on one or more criteria as described herein. Once back in normal operating mode, the TCC **140** may handle a subsequent emergency text from the UE **110** normally. Thus, the TCC **140** will, in response to receiving an emergency text from the UE **110** and represented by the signal path **185**,

determine the current UE location for the UE **110** and the appropriate PSAP for that location, in this example PSAP **170**. The TCC **140** may then forward the emergency text from the UE **110** to the PSAP **170**. The TCC **140** may also, or instead, further facilitate emergency communications between the UE **110** and the PSAP **170**.

(35) In another example, the UE **112** may initiate a voice testing operation by transmitting a voice testing request to the SMSC **130** via the base station **122**. The signal path **182** may represent the path that such a request and related signals may take. The voice testing request from the UE **112** may include a short code (e.g., “933”) that may cause the SMSC **130** to transmit the short code to one or more call routing nodes **150** for transmission to the GMLC **160**. In response to receiving the short code, the GMLC **160** may enter an emergency voice testing mode for the UE **112**. The GMLC **160** may reply with a confirmation message that may include voice testing mode data. The SMSC **130** may forward the reply to the UE **112** with a voice testing mode confirmation communication. This confirmation may indicate a confirmation of voice testing mode at the GMLC **160** for the UE **112** and/or one or more parameters, details, and/or other information associated with the voice testing mode. For example, the confirmation may explicitly indicate that the testing mode will be active for a particular number of test emergency texts, a particular amount of time, until another (same or different) short code is received, etc. The GMLC **160** may store an indicator of voice testing mode associated with an indicator of the UE **112** to maintain a record of voice testing mode for the UE **112**. Here again, the indicator of the UE **112** may take any form, including a telephone number, an MSISDN, an IMSI, an MDN, etc. For example, the GMLC **160** may store an array of identifiers associated with UEs for which the GMLC **160** may be in voice testing mode.

(36) While in voice testing mode for the UE **112**, the GMLC **160** may perform testing operations in response to receiving an emergency voice communication from the UE **112**. The signal path **184** may represent the path that such a test emergency voice call and related signals may take. For example, an operator of the UE **112** may request to initiate a voice call to an emergency number (e.g., 911). This request may be received at the call routing node(s) **150**. The node(s) **150** may transmit a request for call routing information to the GMLC **160** (e.g., in response to determining that the call is an emergency communications). This request may include the location of the UE **112** as included in the request to initiate the call. The GMLC **160** may determine that it is in voice testing mode for the UE **112** based on the UE **112**'s identifier, may determine the UE **112**'s current location (e.g., based on the information received from the nodes(s) **150**), and may determine the appropriate PSAP for the UE **112**'s current location, in this example, PSAP **172**. Instead of setting up the emergency call, the GMLC **160** may reply to the UE **112** with a testing information communication (e.g., a voice communication and/or a text communication) that may indicate the determined PSAP, the determined current UE location, a timestamp, additional PSAP data, and/or other testing information.

(37) The GMLC **160** may exit the voice testing mode or otherwise return to normal text operating mode based on one or more criteria as described herein. Once back in normal operating mode, the GMLC **160** may handle a subsequent emergency voice call from the UE **112** normally. Thus, the GMLC **160** will, in response to receiving a request to establish an emergency call from the UE **112** via call routing node(s) **150** and represented by the signal path **186**, determine the current UE location for the UE **112** and the appropriate PSAP for that location, in this example PSAP **172**. The GMLC **160** may then forward the call routing information for the PSAP **172** to the call routing nodes **150**, which may use this information to establish an emergency call between the UE **112** and PSAP **172**.

(38) Illustrative Signal Flows

(39) FIG. **2** illustrates an exemplary signal flow **200** of various messages that may be exchanged in one or more of the disclosed systems and techniques for more efficiently and accurately testing emergency communications operations using short codes. Reference will be made in this description of the signal flow **200** to devices, entities, and interfaces illustrated in FIG. **1** and

described in regard to that figure. However, the operations, signals, and signal flow illustrated in FIG. **2** and described herein may be implemented in any suitable system and/or with any one or more suitable devices and/or entities. Moreover, any of the operations, signals, and/or entities described in regard to FIG. **2** may be used separately and/or in conjunction with other operations, signals, and/or entities. All such embodiments are contemplated as within the scope of the instant disclosure.

(40) The UE **110** may transmit a test mode initiation request **202** to the TCC **140** (e.g., via one or more gNodeBs, SMSCs, and/or other wireless communications network components). The request **202** may include a UE identifier **204** for the UE **110**. This identifier may be any one or more of a telephone number, an MSISDN, an MDN, an IMSI, etc. The request **202** may also include a short code **206** that may be a code that the TCC **140** may be configured to detect as a request to initiate a text testing mode for a UE indicated in a request that includes such a short code. For example, the request **202** may be a text message with a text body that includes the short code **206** (e.g., “922”) and is associated with an originating device indicated by the UE identifier **204**.

(41) Based on receiving the request **202**, the TCC **140** may determine at operation **208** that the request **202** includes the short code **206** that is associated with a text testing mode. In response to this determination, the TCC **140** may store an indication and/or other data that indicates that the TCC is to perform text testing mode operations for an emergency text received from the UE **110**. For example, the TCC **140** may store an indication of the UE **110** and/or its identifier **204** in a list, array, and/or other data structure that may be associated with text testing mode. Further at **208**, the TCC **140** may set a timer, counter, or other indicator or tracker of text testing mode operations. For example, the TCC **140** may be configured to remain in text testing mode for a particular UE for a particular amount of time (e.g., 1 minute, 5 minutes, etc.) and/or to process a particular number of emergency texts (e.g., 1, 2, 5, 10, etc.) using text testing operations for a particular UE before returning to normal operating mode for that UE. The TCC **140** may also, or instead, be configured to track one or more other criteria associated with text testing mode that may be used to (e.g., automatically) determine if and/or when to revert to normal operating mode for a particular UE. This may be important to ensure that text testing mode does not remain in place indefinitely for a particular UE, preventing legitimate use of that UE for emergency text communications.

(42) The TCC **140** may generate and transmit a text mode notification **210** to the UE **110**. The notification **210** may be a text message. The notification **210** may indicate that the TCC **140** has entered text testing mode for the UE **110**, the time at which the TCC **140** entered text testing mode for the UE **110**, a length of time that the TCC **140** will remain in text testing mode for the UE **110** (e.g., before reverting to normal operation), a number of emergency communications that the TCC **140** will process in text testing mode for the UE **110** (e.g., before reverting to normal operation), an identifier of the TCC **140**, a reiteration of the UE identifier **204**, a reiteration of the short code **206**, and/or any other information that may be associated with the text testing mode and/or the involved components.

(43) The UE **110** may transmit an emergency text **212** to the TCC **140**. The emergency text **212** may include the UE identifier **204** and may also include a UE location **214** for the UE **110**. The UE location **214** (and any other location indicated by a UE described herein) may take a standardized UE location data form (e.g., latitude and longitude) and/or any other form that may enable one or more components of a wireless communications network to determine a current location of a UE.

(44) At operation **216**, the TCC **140** may determine that the emergency text **212** is associated with the UE **110** based on the UE identifier **204**. The TCC **140** may determine that it is in a text testing mode for the UE **110**. Based on this determination, the TCC **140** may determine the UE location **214** based on the emergency text **212**. Using the UE location **214**, the TCC **140** may determine the location of the PSAP associated with that location (e.g., the PSAP that is geographically most proximate, among available PSAPs, to the UE location **214**). The TCC **140** may further determine any other relevant test response data, such as any remaining text testing mode time, remaining

emergency texts to be processed in testing mode, etc. The TCC **140** may transmit the determined PSAP and/or PSAP location, in some examples along with any other test response data, to the UE **110** in test response **218**. Note that in some examples, the test response data determined at operation **216** may also, or instead, be provided to one or more other components and/or systems, such as a database configured to receive and store such testing data.

(45) At operation **220**, the TCC **140** may revert to normal mode for the UE **110** (e.g., for UE identifier **204**). In examples, the TCC **140** may be configured to automatically revert to normal mode after processing one emergency text in text testing mode for the UE **110**. In other examples, the TCC **140** may be configured to automatically revert to normal mode after processing a predetermined number of emergency texts in text testing mode for the UE **110**, after a particular length of time has passed since entering text testing mode for the UE **110**, after receiving another short code from the UE **110** (e.g., a dedicated "stop testing" short code or a repeat of the same testing short code), and/or based on any other criteria, conditions, or combination thereof.

(46) The UE **110** may transmit an emergency text **222** to the TCC **140**. The emergency text **222** may include the UE identifier **204** and may also include a UE location **224** for the UE **110** (e.g., that may be a different location than the UE location **214** as the UE **110** may have changed locations since transmitting the emergency text **212**). At operation **226**, the TCC **140** may determine that the emergency text **222** is associated with the UE **110** based on the UE identifier **204**. The TCC **140** may determine that it is not in a text testing mode for the UE **110**. Based on this determination, the TCC **140** may determine the UE location **224** based on the emergency text **212**. Using the UE location **224**, the TCC **140** may determine the location of the PSAP associated with that location (e.g., the PSAP that is geographically most proximate, among available PSAPs, to the UE location **224**). In this example, the TCC **140** may determine the PSAP **170** is the most proximate PSAP to the UE location **224**. The TCC **140** may then relay or otherwise transmit data associated with the emergency text **222** as emergency message **228** to the PSAP **170**. The PSAP **170** may respond with an emergency response **230** that the TCC **140** may relay or otherwise transmit back to the UE **110** as an emergency response **232**. Additional normal emergency communications operations may be performed.

(47) FIG. **3** illustrates an exemplary signal flow **300** of various messages that may be exchanged in one or more of the disclosed systems and techniques for more efficiently and accurately testing emergency communications operations using short codes. Reference will be made in this description of the signal flow **300** to devices, entities, and interfaces illustrated in FIG. **1** and described in regard to that figure. However, the operations, signals, and signal flow illustrated in FIG. **3** and described herein may be implemented in any suitable system and/or with any one or more suitable devices and/or entities. Moreover, any of the operations, signals, and/or entities described in regard to FIG. **3** may be used separately and/or in conjunction with other operations, signals, and/or entities. All such embodiments are contemplated as within the scope of the instant disclosure.

(48) The UE **112** may transmit a test mode initiation request **302** to the SMSC **130** (e.g., via one or more gNodeBs and/or other wireless communications network components). The request **302** may include a UE identifier **304** for the UE **112**. This identifier may be any one or more of a telephone number, an MSISDN, an MDN, an IMSI, etc. The request **302** may also include a short code **306** that may be a code that represents a request to initiate a voice testing mode for a UE indicated in a request that includes such a short code. For example, the request **302** may be a text message with a text body that includes the short code **306** (e.g., "933") and is associated with an originating device indicated by the UE identifier **304**.

(49) Based on receiving the request **302**, the SMSC **130** may determine that the request **302** includes the short code **306** that is associated with a voice testing mode. In response to this determination, the SMSC **130** may transmit a request **308** to initiate voice testing mode for the UE **112** (as identified by UE identifier **204** in the request **308**) to the GMLC **160**. Alternatively, the

SMSC **130** may relay the request **302** as the request **308** to the GMLC **160**. The request **308** may include the short code **306** and/or any other instructions or data that may be recognized by the GMLC **160** as an instruction to enter a voice testing mode for the UE identified in the request **308**. The GMLC **160**, at operation **310** in response to the request **308**, may store an indication and/or other data that indicates that the GMLC **160** is to perform voice testing mode operations for emergency voice calls received from the UE **112**. For example, the GMLC **160** may store an indication of the UE **112** and/or its identifier **304** in a list, array, and/or other data structure that may be associated with voice testing mode. Further at **310**, the GMLC **160** may set a timer, counter, or other indicator or tracker of voice testing mode operations. For example, the GMLC **160** may be configured to remain in voice testing mode for a particular UE for a particular amount of time (e.g., 1 minute, 5 minutes, etc.) and/or to process a particular number of emergency calls (e.g., 1, 2, 5, 10, etc.) using voice testing operations for a particular UE before returning to normal operating mode for that UE. The GMLC **160** may also, or instead, be configured to track one or more other criteria associated with voice testing mode that may be used to (e.g., automatically) determine if and/or when to revert to normal operating mode for a particular UE. This may be important to ensure that voice testing mode does not remain in place indefinitely for a particular UE, preventing legitimate use of that UE for emergency voice communications.

(50) The GMLC **160** may generate and transmit a voice mode notification **312** to the SMSC **130** that the SMSC **130** may use to generate and transmit an associated voice mode notification **314** to UE **112**. Alternatively, the SMSC **130** may relay the notification **312** as the notification **314** to the UE **112**. Either or both of the notifications **312** and **314** may be a text message. Either or both of the notifications **312** and **314** may indicate that the GMLC **160** has entered voice testing mode for the UE **112**, the time at which the GMLC **160** entered voice testing mode for the UE **112**, a length of time that the GMLC **160** will remain in voice testing mode for the UE **112** (e.g., before reverting to normal operation), a number of emergency voice communications that the GMLC **160** will process in voice testing mode for the UE **112** (e.g., before reverting to normal operation), an identifier of the GMLC **160**, a reiteration of the UE identifier **304**, a reiteration of the short code **306**, and/or any other information that may be associated with the voice testing mode and/or the involved components.

(51) The UE **112** may transmit an emergency call request **316** to the routing node(s) **150** (the TCC **140** may not be involved in voice communications). The emergency call request **316** may include the UE identifier **304** and may also include a UE location **318** for the UE **112**. The UE location **318** (and any other location indicated by a UE described herein) may take a standardized UE location data form (e.g., latitude and longitude) and/or any other form that may enable one or more components of a wireless communications network to determine a current location of a UE.

(52) In response to the call request **316**, the routing node(s) **150** may transmit a request **320** for routing instructions to the GMLC **160**. For example, the routing node(s) **150** may determine, in response to receiving a request to set up an emergency call, that it will need a PSAP location to use as a destination for the call. The routing node(s) **150** may then query the GMLC **160** using the request **320** for a PSAP location (e.g., identifier, network address, geographical location, etc.). The request **320** may include the UE identifier **304** and the UE location **318** and/or other data representative thereof.

(53) At operation **322**, the GMLC **160** may determine that the request **320** for routing instructions is associated with the UE **112** based on the UE identifier **304**. The GMLC **160** may determine that it is in a voice testing mode for the UE **112**. Based on this determination, the GMLC **160** may determine the UE location **318** based on the request **320**. Using the UE location **318**, the GMLC **160** may determine the location of the PSAP associated with that location (e.g., the PSAP that is geographically most proximate, among available PSAPs, to the UE location **318**). The GMLC **160** may further determine any other relevant test response data, such as any remaining voice testing mode time, remaining emergency calls to be processed in testing mode, etc. The GMLC **160** may

transmit the determined PSAP and/or PSAP location, in some examples along with any other test response data, to the routing node(s) **150** as test response **324**. The routing node(s) may forward or otherwise convey this test response data to the UE **112** as test response **326**. In examples, the test responses **324** and **326** may take the form of a voice message (e.g., automatically generated voice message or announcement) that indicates the test response data and that may be played or otherwise presented to the operator of the UE **112**. In other examples, the test responses **324** and **326** may represent operations that may be taken to provide a text message or other type of representation of the test response data to the UE **112**. Note that in some examples, the test response data determined at operation **322** may also, or instead, be provided to one or more other components and/or systems, such as a database configured to receive and store such testing data.

(54) At operation **328**, the GMLC **160** may revert to normal mode for the UE **112** (e.g., for UE identifier **304**). In examples, the GMLC **160** may be configured to automatically revert to normal mode after processing one emergency call in voice testing mode for the UE **112**. In other examples, the GMLC **160** may be configured to automatically revert to normal mode after processing a predetermined number of emergency calls in voice testing mode for the UE **112**, after a particular length of time has passed since entering voice testing mode for the UE **112**, after receiving another short code from the UE **112** (e.g., a dedicated "stop testing" short code or a repeat of the same testing short code), and/or based on any other criteria, conditions, or combination thereof.

(55) The UE **112** may transmit an emergency call request **330** to the routing node(s) **150**. The emergency call request **330** may include the UE identifier **304** and may also include a UE location **332** for the UE **112** (e.g., that may be a different location than the UE location **318** as the UE **112** may have changed locations since transmitting the emergency text **212**). In response to the call request **330**, the routing node(s) **150** may transmit a request **334** for routing instructions to the GMLC **160**. For example, the routing node(s) **150** may determine, in response to receiving a request to set up an emergency call, that it will need a PSAP location to use as a destination for the call. The routing node(s) **150** may then query the GMLC **160** using the request **334** for a PSAP location (e.g., identifier, network address, geographical location, etc.). The request **320** may include the UE identifier **304** and the UE location **332** and/or other data representative thereof.

(56) At operation **336**, the GMLC **160** may determine that the request **334** for routing instructions is associated with the UE **112** based on the UE identifier **304**. The GMLC **160** may determine that it is not in a voice testing mode for the UE **112**. Based on this determination, the GMLC **160** may determine the UE location **318** based on the request **334**. Using the UE location **332**, the GMLC **160** may determine the location of the PSAP associated with that UE location (e.g., the PSAP that is geographically most proximate, among available PSAPs, to the UE location **332**). In this example, the GMLC **160** may determine that the PSAP **172** is the most proximate PSAP the location indicated by UE location **332**. The GMLC **160**, at operation **336**, may further determine any other relevant routing instruction data and/or any other emergency communications data that may be used for establishing an emergency call between the UE **112** and the PSAP **172**. The GMLC **160** may transmit the determined PSAP and/or PSAP location, in some examples along with any other test response data, to the routing node(s) **150** as emergency routing data **338**. The routing node(s) use the emergency routing data **338** to transmit a request **340** to the PSAP **172** to establish an emergency call with the UE **112**. The emergency call **342** may then be established between the PSAP **172** and the UE **112**. Additional normal emergency communications operations may be performed.

(57) Illustrative Operations

(58) FIG. **4** shows a flow diagram of an illustrative process **400** for performing emergency communications testing using short codes according to the disclosed embodiments. The process **400** is illustrated as a collection of blocks in a logical flow diagram, which represents a sequence of operations that can be implemented in software and executed in hardware. In the context of software, the blocks represent computer-executable instructions that, when executed by one or

more processors, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform functions and/or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks can be omitted and/or combined in any order and/or in parallel to implement the processes. For discussion purposes, the process **400** may be described with reference to the wireless network environment **100** of FIG. **1**; however other environments may also be used.

(59) At block **402**, a component of a wireless communications network, such as a TCC, may receive a text testing mode initiation request from a UE. This request may include a particular text short code that may be associated with text testing mode. As noted, such short codes may take any form (e.g., “922”). This request may also include an identifier of the originating UE.

(60) At block **404**, the TCC may determine one or more text testing mode parameters. For example, the TCC may determine a number of texts to be processed before reverting to normal mode, a length of time to remain in text testing mode before (e.g., automatically) reverting to normal mode, etc. Further at block **404**, the TCC may enter a text testing mode for communications originating with the UE that sent the text testing mode short code received at **402**. The TCC may store an indication of the UE in a data structure associated with text testing mode as described herein and/or may set a counter or timer as described herein to track text testing mode operations (e.g., to determine when to exit text testing mode).

(61) At block **406**, the TCC may reply to the originating UE with a testing mode confirmation communication that may include one or more parameters, details, and/or other information associated with the text testing mode. For example, the TCC may reply with a text message that indicates that text testing mode has begun for the requesting UE and a number of processed emergency texts and/or a length of time until the TCC will revert to normal operation.

(62) At block **408**, the TCC may receive an emergency text from a UE. The TCC may determine, at block **410**, if it is in text testing mode for the UE associated with the emergency text. For example, the TCC may determine whether the emergency text was received from the UE that initially transmitted the text testing mode initiation request received at block **402**. If the TCC is not in text testing mode for the UE associated with the emergency text, the process may move to block **420** for the TCC to process the emergency text normally.

(63) If, at block **410**, the TCC determines that it is in text testing mode for the UE associated with the emergency text received at **408**, at block **412** the TCC may determine text test response data based on the UE and the emergency text. For example, the TCC may determine the UE's current location (e.g., as indicated in the emergency text). The TCC may then determine the appropriate PSAP for the UE's current location. The TCC may also, or instead, determine other responsive testing data. At block **414**, the TCC may transmit this responsive data (e.g., in a text message) to the UE associated with the received emergency text. In this responsive message, the TCC may also indicate other information as described herein (e.g., the determined current UE location, a timestamp, additional PSAP data, remaining testing texts to process, remaining time in test testing mode, etc.).

(64) At block **416** the TCC may determine whether to exit text testing mode for the UE associated with the emergency text received at **408**. In various examples, the TCC may exit the text testing mode or otherwise return to normal text operating mode after responding to a threshold number (e.g., 1, 2, 3, 20, etc.) of emergency texts. Alternatively or additionally, the TCC may exit the text testing mode or otherwise return to normal text operating mode after a threshold period of time (e.g., 5 minutes, 10 minutes, 1 hour, etc.) has passed, in some examples, regardless of the number of emergency texts processed in text testing mode. Alternatively or additionally, the TCC may exit the text testing mode or otherwise return to normal text operating mode after receiving another short code from the originating UE. Any other criteria may be used to determine whether to continue in a text testing mode for a particular UE by a TCC.

(65) If the TCC determines to remain in text testing mode for the particular UE at block **416**, the process **400** may return to block **408** to receive additional emergency texts for processing. If the TCC determines to exit text testing mode for the particular UE at block **416**, the process **400** may proceed to block **418** where the TCC may set the particular UE to normal mode for future emergency text processing operations. The process **400** may return to block **408** to receive additional emergency texts for processing.
(66) If the TCC is not in text testing mode and has received an emergency text, for example, at block **408**, the process **400** may proceed to block **420** where the TCC may determine the UE's location and the appropriate PSAP for that location. The TCC may, at block **422** and in response to receiving the emergency text and determining the appropriate PSAP, facilitate emergency text communications between the UE and the PSAP.
(67) FIG. **5** shows a flow diagram of an illustrative process **500** for performing emergency communications testing using short codes according to the disclosed embodiments. The process **500** is illustrated as a collection of blocks in a logical flow diagram, which represents a sequence of operations that can be implemented in software and executed in hardware. In the context of software, the blocks represent computer-executable instructions that, when executed by one or more processors, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform functions and/or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks can be omitted and/or combined in any order and/or in parallel to implement the processes. For discussion purposes, the process **500** may be described with reference to the wireless network environment **100** of FIG. **1**; however other environments may also be used.
(68) At block **502**, a component of a wireless communications network, such as an SMSC, may receive a voice testing mode initiation request from a UE. This request may include a particular text short code that may be associated with voice testing mode. As noted, such short codes may take any form (e.g., “933”). This request may also include an identifier of the originating UE.
(69) At block **504**, the SMSC may transmit a request to a GMLC to initiate voice testing mode for the UE associated with the request received at block **502**. This request to the GMLC may include the UE identifier and an instruction to initiate the voice testing mode. For example, this request may be simply a relay of the request received at block **502**. Alternatively, this request may be a request generated by the SMSC based on the request received at block **502**. The request to the GMLC may include the short code received at block **502** and/or instruction of another form that may cause the GMLC to enter voice testing mode for the UE.
(70) At block **506**, the SMSC may receive a confirmation of voice testing mode from the GMLC. This confirmation may include other information, such as a length of time the GMLC will remain in voice testing mode for the UE, a number of calls the GMLC will process in testing mode, a timestamp when the GMLC entered testing mode for the UE, etc.
(71) At block **508**, the SMSC may reply back to the originating UE with a voice testing mode confirmation communication that may include one or more parameters, details, and/or other information associated with the voice testing mode, such as those received from the GMLC at block **506**. For example, the SMSC may generate a text message that indicates that voice testing mode has begun for the requesting UE at the GMLC and a number of processed emergency calls and/or a length of time until the GMLC will revert to normal operation. Alternatively, the SMSC may relay the confirmation message received from the GMLC at block **506**.
(72) FIG. **6** shows a flow diagram of an illustrative process **600** for performing emergency communications testing using short codes according to the disclosed embodiments. The process **600** is illustrated as a collection of blocks in a logical flow diagram, which represents a sequence of operations that can be implemented in software and executed in hardware. In the context of software, the blocks represent computer-executable instructions that, when executed by one or

more processors, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform functions and/or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks can be omitted and/or combined in any order and/or in parallel to implement the processes. For discussion purposes, the process **600** may be described with reference to the wireless network environment **100** of FIG. **1**; however other environments may also be used.
(73) At block **602**, a component of a wireless communications network, such as a GMLC, may receive a voice testing mode initiation request from an SMSC (e.g., originating at a UE and generated, transmitted and/or relayed in process **500** of FIG. **5**). This request may include an instruction to begin voice testing mode for a particular UE. For example, the request may include a particular short code that may be associated with voice testing mode (or any other form of voice testing mode instruction) and an identifier of the UE.
(74) At block **604**, the GMLC may determine one or more voice testing mode parameters. For example, the GMLC may determine a number of calls to be processed before reverting to normal mode, a length of time to remain in voice testing mode before (e.g., automatically) reverting to normal mode, etc. Further at block **604**, the GMLC may enter a voice testing mode for communications originating with the UE indicated in the voice testing mode initiation request received at **602**. The GMLC may store an indication of the UE in a data structure associated with voice testing mode as described herein and/or may set a counter or timer as described herein to track voice testing mode operations (e.g., to determine when to exit voice testing mode).
(75) At block **606**, the GMLC may reply back to the SMSC with a testing mode confirmation communication that may include one or more parameters, details, and/or other information associated with the voice testing mode. For example, the GMLC may reply with a text message or text message content that indicates that text testing mode has begun for the requesting UE and a number of processed emergency calls and/or a length of time until the GMLC will revert to normal operation. The recipient SMSC may relay this message to the UE or use this data to generate a (e.g., text) notification to the requesting UE.
(76) At block **608**, the GMLC may receive a request from one or more routing nodes for routing instructions for an emergency call originating at a UE. This request may include an identifier of the UE requesting establishment of the emergency call and location data for that UE. The GMLC may determine, at block **610**, if it is in voice testing mode for the UE associated with the emergency call. For example, the GMLC may determine whether the emergency call request was received from the UE that initially transmitted the voice testing mode initiation request that triggered the request received at block **602**. If the GMLC is not in voice testing mode for the UE associated with the emergency call request, the process may move to block **620** for the GMLC to process the request for emergency call routing information normally.
(77) If, at block **610**, the GMLC determines that it is in voice testing mode for the UE associated with the emergency call for which routing instructions have been requested at **608**, at block **612** the GMLC may determine voice test response data based on the UE and the emergency call routing instructions request. For example, the GMLC may determine the UE's current location (e.g., as indicated in the emergency call routing instructions request). The GMLC may then determine the appropriate PSAP for the UE's current location. The GMLC may also, or instead, determine other responsive testing data. At block **614**, the GMLC may transmit this responsive data (e.g., as a voice message or announcement) to the UE associated with the requested emergency call (e.g., via one or more routing nodes). In this responsive message, the GMLC may also indicate other information as described herein (e.g., the determined current UE location, a timestamp, additional PSAP data, remaining testing calls to process, remaining time in voice testing mode, etc.).
(78) At block **616** the GMLC may determine whether to exit voice testing mode for the UE associated with the emergency calling routing instructions request received at **608**. In various

examples, the GMLC may exit the voice testing mode or otherwise return to normal voice operating mode after responding to a threshold number (e.g., 1, 2, 3, 20, etc.) of requests for emergency call routing instructions. Alternatively or additionally, the GMLC may exit the voice testing mode or otherwise return to normal voice operating mode after a threshold period of time (e.g., 5 minutes, 10 minutes, 1 hour, etc.) has passed, in some examples, regardless of the number of requests for emergency call routing instructions processed in voice testing mode. Alternatively or additionally, the GMLC may exit the voice testing mode or otherwise return to normal voice operating mode after receiving another instruction from a UE via an SMSC. Any other criteria may be used to determine whether to continue in a voice testing mode for a particular UE by a GMLC.
(79) If the GMLC determines to remain in voice testing mode for the particular UE at block **616**, the process **600** may return to block **608** to receive additional requests for emergency call routing instructions for processing. If the GMLC determines to exit voice testing mode for the particular UE at block **616**, the process **600** may proceed to block **618** where the GMLC may set the particular UE to normal mode for future requests for emergency call routing instruction request processing operations. The process **600** may return to block **608** to receive and process additional requests for emergency call routing instructions.
(80) If the GMLC is not in voice testing mode and has received a request for emergency call routing instructions, for example, at block **608**, the process **600** may proceed to block **620** where the GMLC may determine the UE's location and the appropriate PSAP for that location. The GMLC may, at block **622** and in response to receiving the routing instructions request and determining the appropriate PSAP, facilitate emergency voice communications between the UE and the PSAP, for example, by providing the routing instructions to one or more call routing nodes so that an emergency call may be established.
(81) In summary, by more efficiently and accurately testing emergency communications operations and components without disrupting operational emergency services systems and personnel, the disclosed systems and techniques may be able to increase the efficiency of usage of emergency resources and wireless network resources, improving the user experience and performance of both the network and user devices.
(82) Example User Equipment
(83) FIG. **7** is an example of a UE, such as UE **110** or UE **112**, for use with the systems and methods disclosed herein, in accordance with some examples of the present disclosure. The UE **110**/**112** may include one or more processors **702**, one or more transmit/receive antennas (e.g., transceivers or transceiver antennas) **704**, and a data storage **706**. The data storage **706** may include a computer readable media **708** in the form of memory and/or cache. This computer-readable media may include a non-transitory computer-readable media. The processor(s) **702** may be configured to execute instructions, which can be stored in the computer readable media **708** and/or in other computer readable media accessible to the processor(s) **702**. In some configurations, the processor(s) **702** is a Central Processing Unit (CPU), a Graphics Processing Unit (GPU), or both CPU and GPU, or any other sort of processing unit. The transceiver antenna(s) **704** can exchange signals with a base station, such as base stations **120** and **122**.
(84) The UE **110**/**112** may be configured with a memory **710**. The memory **710** may be implemented within, or separate from, the data storage **706** and/or the computer readable media **708**. The memory **710** may include any available physical media accessible by a computing device to implement the instructions stored thereon. For example, the memory **710** may include, but is not limited to, RAM, ROM, EEPROM, a SIM card, flash memory or other memory technology, CD-ROM, DVD or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by the UE **110**/**112**.
(85) The memory **710** can store several modules, such as instructions, data stores, and so forth that are configured to execute on the processor(s) **702**. In configurations, the memory **710** may also

store one or more applications **714** configured to receive and/or provide voice, data, and messages (e.g., SMS messages, Multi-Media Message Service (MMS) messages, Instant Messaging (IM) messages, Enhanced Message Service (EMS) messages, etc.) to and/or from another device or component (e.g., the base stations **120** and **122**). The applications **714** may also include one or more operating systems and/or one or more third-party applications that provide additional functionality to the UE **110**/**112**.

(86) Although not all illustrated in FIG. **8**, the UE **110**/**112** may also comprise various other components, e.g., a battery, a charging unit, one or more network interfaces **716**, an audio interface, a display **718**, a keypad or keyboard, and one or more input devices **720**, and one or more output devices **722**. The UE **110**/**112** may further include one or more location determination components **724** that may be configured to determine a current UE location that may be used as described herein for testing emergency communications operations and components.

(87) Example Computing Device

(88) FIG. **8** is an example of a computing device **800** for use with the systems and methods disclosed herein, in accordance with some examples of the present disclosure. The computing device **800** can be used to implement various components of a core network, a base station, and/or any servers, routers, gateways, gateway elements, administrative components, etc. that can be used by a communication provider. One or more computing devices **800** can be used to implement the network **101**, for example. One or more computing devices **800** can also be used to implement base stations and other components.

(89) In various embodiments, the computing device **800** can include one or more processing units **802** and system memory **804**. Depending on the exact configuration and type of computing device, the system memory **804** can be volatile (such as RAM), nonvolatile (such as ROM, flash memory, etc.) or some combination of the two. The system memory **804** can include an operating system **806**, one or more program modules **808**, program data **810**, and one or more digital certificates **820**. The system memory **804** may be secure storage or at least a portion of the system memory **804** can include secure storage. The secure storage can prevent unauthorized access to data stored in the secure storage. For example, data stored in the secure storage can be encrypted or accessed via a security key and/or password.

(90) The computing device **800** can also include additional data storage devices (removable and/or non-removable) such as, for example, magnetic disks, optical disks, or tape. Such additional storage is illustrated in FIG. **8** by storage **812**.

(91) Non-transitory computer storage media of the computing device **800** can include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. The system memory **804** and storage **812** are examples of computer readable storage media. Non-transitory computer readable storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile discs (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computing device **800**. Any such non-transitory computer readable storage media can be part of the computing device **800**.

(92) In various embodiment, any or all of the system memory **804** and storage **812** can store programming instructions which, when executed, implement some or all of the functionality described above as being implemented by one or more systems configured in the environment **100** and/or components of the network **101**.

(93) The computing device **800** can also have one or more input devices **814** such as a keyboard, a mouse, a touch-sensitive display, voice input device, etc. The computing device **800** can also have one or more output devices **816** such as a display, speakers, a printer, etc. can also be included. The computing device **800** can also contain one or more communication connections **818** that allow the

device to communicate with other computing devices using wired and/or wireless communications.

EXAMPLE CLAUSES

(94) The following paragraphs describe various examples. Any of the examples in this section may be used with any other of the examples in this section and/or any of the other examples or embodiments described herein. A: A method performed by a text control center (TCC), the method comprising: receiving, at the TCC from a user equipment (UE), a first text message comprising a short code associated with an emergency communications testing mode; initiating, at the TCC based at least in part on the short code, the emergency communications testing mode for the UE at the TCC; receiving, at the TCC from the UE, an emergency text message; determining, at the TCC, that the TCC is in the emergency communications testing mode for the UE; based at least in part on determining that the TCC is in the emergency communications testing mode for the UE: determining a location for the UE based at least in part on the emergency text message; determining a public safety access point (PSAP) based at least in part on the location; and transmitting, from the TCC to the UE, a second text message comprising an indication of the PSAP; and terminating, at the TCC, the emergency communications testing mode for the UE. B: The method of paragraph A, further comprising: receiving, at the TCC from the UE, a second emergency text message; determining, at the TCC, that the TCC is not in the emergency communications testing mode for the UE; and based at least in part on determining that the TCC is not in the emergency communications testing mode for the UE: determining a second location for the UE based at least in part on the second emergency text message; determining a second PSAP based at least in part on the second location; and transmitting, from the TCC to the second PSAP, the second emergency text message. C: The method of paragraph A or B, wherein terminating the emergency communications testing mode for the UE at the TCC is based at least in part on transmitting the second text message. D: The method of any of paragraphs A-C, wherein terminating the emergency communications testing mode for the UE at the TCC is based at least in part on determining that a threshold amount of time has passed since initiating the emergency communications testing mode. E: The method of any of paragraphs A-D, wherein terminating the emergency communications testing mode for the UE at the TCC is based at least in part on receiving a second short code associated with terminating the emergency communications testing mode. F: The method of any of paragraphs A-E, wherein determining the location for the UE comprises determining a latitude and a longitude for the UE represented in the emergency text message. G: A gateway mobile location center (GMLC) comprising: one or more processors; one or more transceivers; and non-transitory computer-readable media storing computer-executable instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising: receiving, at the GMLC from a text control center (TCC), an instruction to initiate an emergency communications testing mode for a user equipment (UE) at the GMLC; initiating the emergency communications testing mode for the UE at the GMLC; receiving, at the GMLC from a call routing node, a request for routing instructions for an emergency call requested by the UE; determining, at the GMLC, that the GMLC is in the emergency communications testing mode for the UE; based at least in part on determining that the GMLC is in the emergency communications testing mode for the UE: determining a location for the UE based at least in part on the request for the routing instructions; determining a public safety access point (PSAP) based at least in part on the location; and transmitting, from the GMLC to the UE, a message comprising an indication of the PSAP; and terminating the emergency communications testing mode for the UE at the GMLC. H: The GMLC of paragraph G, wherein the instruction to initiate the emergency communications testing mode comprises a short code associated with the emergency communications testing mode. I: The GMLC of paragraph G or H, wherein the message comprising the indication of the PSAP comprises a voice message comprising the indication of the PSAP. J: The GMLC of any of paragraphs G-I, wherein initiating the emergency communications testing mode comprises starting a timer at the GMLC. K: The GMLC of any of paragraphs G-J,

wherein terminating the emergency communications testing mode for the UE comprises: determining, based at least in part on the timer, that a threshold amount of time has passed since initiating the emergency communications testing mode; and terminating the emergency communications testing mode for the UE based at least in part on determining that the threshold amount of time has passed. L: The GMLC of any of paragraphs G-K, wherein the operations further comprise: receiving, at the GMLC from a second call routing node, a second request for second routing instructions for a second emergency call requested by the UE; determining that the GMLC is not in the emergency communications testing mode for the UE; and based at least in part on determining that the GMLC is not in the emergency communications testing mode for the UE: determining a second location for the UE based at least in part on the second request for routing instructions; determining a second PSAP based at least in part on the second location; and transmitting, from the GMLC to the second call routing node, the second routing instructions comprising an indication of the second PSAP. M: The GMLC of any of paragraphs G-L, wherein terminating the emergency communications testing mode for the UE is based at least in part on transmitting the message comprising the indication of the PSAP. N: The GMLC of any of paragraphs G-M, wherein: the operations further comprise receiving a second instruction to terminate the emergency communications testing mode for the UE at the GMLC; and terminating the emergency communications testing mode for the UE is based at least in part on the second instruction. O: A non-transitory computer-readable media storing computer-executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving, at a wireless communications network component from a user equipment (UE), a text message comprising a short code associated with an emergency communications testing mode; initiating, at the wireless communications network component based at least in part on the short code, the emergency communications testing mode for the UE; receiving emergency communications from the UE; determining that the UE is associated with the emergency communications testing mode; based at least in part on determining that the UE is associated with the emergency communications testing mode: determining a location for the UE based at least in part on the emergency communications; determining a public safety access point (PSAP) based at least in part on the location; and transmitting an indication of the PSAP to the UE; and terminating the emergency communications testing mode for the UE. P: The non-transitory computer-readable media of paragraph O, wherein initiating the emergency communications testing mode for the UE comprises transmitting an instruction to a gateway mobile location center (GMLC) to initiate the emergency communications testing mode for the UE at the GMLC. Q: The non-transitory computer-readable media of paragraph O or P, wherein terminating the emergency communications testing mode for the UE comprises: receiving a second text message comprising the short code; and terminating the emergency communications testing mode for the UE based at least in part on receiving the second text message. R: The non-transitory computer-readable media of any of paragraphs O-Q, wherein determining the location for the UE comprises determining a latitude and longitude for the UE from the emergency communications. S: The non-transitory computer-readable media of any of paragraphs O-R, wherein initiating the emergency communications testing mode for the UE comprises: determining an identifier for the UE from the text message; and associating the identifier for the UE with the emergency communications testing mode. T: The non-transitory computer-readable media of any of paragraphs O-S, wherein terminating the emergency communications testing mode for the UE is based at least in part on determining that a threshold amount of time has passed since initiating the emergency communications testing mode.

(95) While the example clauses described above are described with respect to one particular implementation, it should be understood that, in the context of this document, the content of the example clauses can also be implemented via a method, device, system, computer-readable medium, and/or another implementation. Additionally, any of the examples A-T can be

implemented alone or in combination with any other one or more of the examples A-T.

CONCLUSION

(96) Depending on the embodiment, certain operations, acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the algorithm). Moreover, in certain embodiments, acts or events can be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially.

(97) The various illustrative logical blocks, components, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

(98) The various illustrative logical blocks, modules, and components described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

(99) The elements of a method, process, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of computer-readable storage medium known in the art. An exemplary storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The processor and the storage medium can reside in an ASIC. The ASIC can reside in a user terminal. In the alternative, the processor and the storage medium can reside as discrete components in a user terminal.

(100) Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or states. Thus, such conditional language is not generally intended to imply that features, elements, and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” “involving,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

(101) Unless otherwise explicitly stated, articles such as “a” or “the” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.
(102) While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, certain embodiments of the inventions described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of certain inventions disclosed herein is indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.
(103) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the claims.

Claims

1. A method performed by a text control center (TCC), the method comprising: receiving, at the TCC from a user equipment (UE), a first text message comprising a short code associated with an emergency communications testing mode; initiating, at the TCC based at least in part on the short code, the emergency communications testing mode for the UE at the TCC; receiving, at the TCC from the UE, an emergency text message; determining, at the TCC, that the TCC is in the emergency communications testing mode for the UE; based at least in part on determining that the TCC is in the emergency communications testing mode for the UE: determining a location for the UE based at least in part on the emergency text message; determining a public safety access point (PSAP) based at least in part on the location; and transmitting, from the TCC to the UE, a second text message comprising an indication of the PSAP; and terminating, at the TCC, the emergency communications testing mode for the UE.
2. The method of claim 1, further comprising: receiving, at the TCC from the UE, a second emergency text message; determining, at the TCC, that the TCC is not in the emergency communications testing mode for the UE; and based at least in part on determining that the TCC is not in the emergency communications testing mode for the UE: determining a second location for the UE based at least in part on the second emergency text message; determining a second PSAP based at least in part on the second location; and transmitting, from the TCC to the second PSAP, the second emergency text message.
3. The method of claim 1, wherein terminating the emergency communications testing mode for the UE at the TCC is based at least in part on transmitting the second text message.
4. The method of claim 1, wherein terminating the emergency communications testing mode for the UE at the TCC is based at least in part on determining that a threshold amount of time has passed since initiating the emergency communications testing mode.
5. The method of claim 1, wherein terminating the emergency communications testing mode for the UE at the TCC is based at least in part on receiving a second short code associated with terminating the emergency communications testing mode.
6. The method of claim 1, wherein determining the location for the UE comprises determining a

latitude and a longitude for the UE represented in the emergency text message.
7. A gateway mobile location center (GMLC) comprising: one or more processors; one or more transceivers; and non-transitory computer-readable media storing computer-executable instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising: receiving, at the GMLC from a text control center (TCC), an instruction to initiate an emergency communications testing mode for a user equipment (UE) at the GMLC; initiating the emergency communications testing mode for the UE at the GMLC; receiving, at the GMLC from a call routing node, a request for routing instructions for an emergency call requested by the UE; determining, at the GMLC, that the GMLC is in the emergency communications testing mode for the UE; based at least in part on determining that the GMLC is in the emergency communications testing mode for the UE: determining a location for the UE based at least in part on the request for the routing instructions; determining a public safety access point (PSAP) based at least in part on the location; and transmitting, from the GMLC to the UE, a message comprising an indication of the PSAP; and terminating the emergency communications testing mode for the UE at the GMLC.
8. The GMLC of claim 7, wherein the instruction to initiate the emergency communications testing mode comprises a short code associated with the emergency communications testing mode.
9. The GMLC of claim 7, wherein the message comprising the indication of the PSAP comprises a voice message comprising the indication of the PSAP.
10. The GMLC of claim 7, wherein initiating the emergency communications testing mode comprises starting a timer at the GMLC.
11. The GMLC of claim 10, wherein terminating the emergency communications testing mode for the UE comprises: determining, based at least in part on the timer, that a threshold amount of time has passed since initiating the emergency communications testing mode; and terminating the emergency communications testing mode for the UE based at least in part on determining that the threshold amount of time has passed.
12. The GMLC of claim 7, wherein the operations further comprise: receiving, at the GMLC from a second call routing node, a second request for second routing instructions for a second emergency call requested by the UE; determining that the GMLC is not in the emergency communications testing mode for the UE; and based at least in part on determining that the GMLC is not in the emergency communications testing mode for the UE: determining a second location for the UE based at least in part on the second request for routing instructions; determining a second PSAP based at least in part on the second location; and transmitting, from the GMLC to the second call routing node, the second routing instructions comprising an indication of the second PSAP.
13. The GMLC of claim 7, wherein terminating the emergency communications testing mode for the UE is based at least in part on transmitting the message comprising the indication of the PSAP.
14. The GMLC of claim 7, wherein: the operations further comprise receiving a second instruction to terminate the emergency communications testing mode for the UE at the GMLC; and terminating the emergency communications testing mode for the UE is based at least in part on the second instruction.
15. A non-transitory computer-readable media storing computer-executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving, at a wireless communications network component from a user equipment (UE), a text message comprising a short code associated with an emergency communications testing mode; initiating, at the wireless communications network component based at least in part on the short code, the emergency communications testing mode for the UE; receiving emergency communications from the UE; determining that the UE is associated with the emergency communications testing mode; based at least in part on determining that the UE is associated with the emergency communications testing mode: determining a location for the UE based at least in part on the emergency communications; determining a public safety access point (PSAP) based at

least in part on the location; and transmitting an indication of the PSAP to the UE; and terminating the emergency communications testing mode for the UE.

16. The non-transitory computer-readable media of claim 15, wherein initiating the emergency communications testing mode for the UE comprises transmitting an instruction to a gateway mobile location center (GMLC) to initiate the emergency communications testing mode for the UE at the GMLC.

17. The non-transitory computer-readable media of claim 15, wherein terminating the emergency communications testing mode for the UE comprises: receiving a second text message comprising the short code; and terminating the emergency communications testing mode for the UE based at least in part on receiving the second text message.

18. The non-transitory computer-readable media of claim 15, wherein determining the location for the UE comprises determining a latitude and longitude for the UE from the emergency communications.

19. The non-transitory computer-readable media of claim 15, wherein initiating the emergency communications testing mode for the UE comprises: determining an identifier for the UE from the text message; and associating the identifier for the UE with the emergency communications testing mode.

20. The non-transitory computer-readable media of claim 15, wherein terminating the emergency communications testing mode for the UE is based at least in part on determining that a threshold amount of time has passed since initiating the emergency communications testing mode.